

# Milestone 4

## 1. Dataset preparation

The required libraries like Pandas, NumPy, and Matplotlib are imported for data handling, computation, and visualization. The dataset is loaded using `read_csv()` to bring structured data into a DataFrame for analysis. The `df.columns` function is used to view and understand the dataset structure. The columns are then renamed to make them simple, consistent, and easy to use in coding. This prepared dataset (from previous milestones) is now ready for further analysis and visualization.

## 2. Incident Type Distribution Analysis

The `value_counts()` function is used to count the frequency of each disaster type in the incident Type column, helping identify how often each event occurs. The result shows that storms, floods, and hurricanes are the most common disasters. The bar plot using Matplotlib is created to visually compare these frequencies for better understanding. This code is used to analyze disaster distribution patterns, which helps in identifying dominant disaster types. The output includes both a numerical frequency table and a graphical representation (bar chart).

## 3. State vs Incident Type

The `groupby()` function is used to group data by state and incident type, and `size()` counts occurrences, creating a matrix of disaster frequencies. The `unstack()` and `fillna(0)` functions convert it into a table format suitable for analysis and visualization. A stacked bar chart using Matplotlib shows how different disasters are distributed across states. The heatmap using Seaborn highlights intensity patterns, making it easier to identify disaster-prone regions. The output reveals that different states experience different types of disasters based on geography.

## 4. Disaster Assistance Program Analysis

The map() function is used to convert 'Yes' and 'No' values into numerical form (1 and 0) for easier analysis. The groupby() function then aggregates the total number of times each disaster type triggered IH and PA assistance programs. A bar chart is plotted using Matplotlib to compare assistance across disaster types. This code is used to understand the impact of disasters on individuals and public infrastructure. The output shows that major disasters like storms, floods, and hurricanes require more assistance, especially public assistance programs.

## CONCLUSION

**The analysis concludes that major disaster types like storms, floods, and hurricanes have the highest impact, as they frequently require both Individual (IH) and Public (PA) assistance programs. Public assistance is used more often, indicating greater damage to infrastructure and public facilities. This shows that large-scale disasters not only affect individuals but also heavily impact government resources. Overall, disaster management efforts are more concentrated on high-frequency and high-impact events.**

**Link of the dashboard:- <https://disaster-monitor-sigma.vercel.app>**

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